COURSE PROGRAMME

1. Information about the programme

1.1 University	**University "Alexandru Ioan Cuza" of Iasi**
1.2 Faculty	**Faculty of Computer Science**
1.3 Department	**Department of Computer Science**
1.4 Domain	**Computer Science**
1.5 Cycle	**Licence**
1.6 Programme / Qualification	**Computer Science (in English)**

2. Information about the course

2.1 Course Name			Numerical Calculus				
2.2 Course taught by			Assoc. Prof. PhD. ANDREEA-VALENTINA ARUSOAIE/ Assoc. Prof. PhD. ANCA IGNAT/ Assoc. Prof. PhD. ANCA IGNAT				
2.3 Seminary / laboratory taught by			Assoc. Prof. PhD. ANDREEA-VALENTINA ARUSOAIE/ Assoc. Prof. PhD. ANCA IGNAT/ Assoc. Prof. PhD. ANCA IGNAT/ Specialist PhD. SEBASTIAN-ADRIAN CIOBANU/ Specialist ROBERT-ADRIAN GAINA/ PhD Student RADU MIRON/ PhD Student IULIA-CATALINA PLESCA				
2.4 Year	III	**2.5** Semester	VI	**2.6** Type of evaluation*	V	**2.7** Course type**	Ob

**E – Exam / C – Colloquium / V – Verification*

***OB – Obligatory / OP – Optionally / F – Facultative*

3. Total hours (estimated per semester and activities)

3.1 Number of hours per week	4	**3.2** course	2	**3.3** seminary/laboratory	2
3.4 Total number of hours	48	**3.5** course	24	**3.6** seminary/laboratory	24

Distribution	hours
Individual study using textbooks, course notes, bibliography items, etc.	14
Supplimentary study (library, on-line platforms, etc.)	7
Individual study for seminary/laboratory, homeworks, projects, etc.	21
Tutoring	0
Examination	10
Other activities	0

3.7 Total hours of individual activity*	52
3.8 Total hours per semester	100
3.9 Credit points	4

4. Pre-requisites - Curriculum (if necessary)

Programming course
Mathematics - differential and integral calculus

5. Conditions (if necessary)

5.1 Course	Videoprojector
5.2 Seminary / Laboratory	Internet

6. Objectives

Acquiring basic knowledge/skills in the field of Numerical Calculus.
Knowing the concepts of linear algebra, interpolation, nonlinear equations, and unconstrained numerical optimization.
Knowing the principles of developing applications that need numerical methods for obtaining the solution
Knowledge of the methods of applying the theoretical knowledge of Numerical Calculation.
Knowledge of the methods of using various Numerical Calculation libraries.

7. Specific competencies/Learning outcomes

- Capacity for analysis and synthesis;
- Ability for self-directed learning aimed at professional development in the chosen field of specialization;
- Critical analysis of specialized literature;
- Use of basic concepts to understand more complex ideas;
- Ability to use mathematical tools for algorithm analysis.

8. Contents

8.1 Course	**Teaching methods**	**Remarks** (number oh hours, references)
Examples, floating point computing, types of errors, propagation of errors, vector and matrix computations, special type matrices	Exposition, debate, case studies, exercises	2, [1,4]
Norms (vectorial, matricial). History of linear systems of equations solving	Exposition, debate, case studies, exercises	2, [1,2]
Errors in numerical computations. Conditioning, stability	Exposition, debate, case studies, exercises	2, [1,2,4]
Substitution methods for triangular systems of equations. Gauss elimination algorithm	Exposition, debate, case studies, exercises	2, [1,2]
LU Decompositions	Exposition, debate, case studies, exercises	2, [1,2]
QR Decompositions	Exposition, debate, case studies, exercises	2, [1,2]
Storing sparse matrices. Iterative methods for solving sparse systems of equations: Jacobi and Gauss-Seidel methods	Exposition, debate, case studies, exercises	2, [1,2,3]
Solving linear system of equations: recapitulation, examples	Exposition, debate, case studies, exercises	2, [1,2]
Iterative methods for symmetric and positive definite matrices: SOR, steapest descent, conjugate gradients	Exposition, debate, case studies, exercises	2, [1,2,3]
Methods for solving the eigenproblem: QR type methods. Methods for approximating the eigenvalues/eigenvectors: power and inverse iteration methods, Hessenberg form Singular Value Decomposition	Exposition, debate, case studies, exercises	2, [1,2]

8.1 Course	Teaching methods	Remarks (number oh hours, references)
Nonlinear equations roots: bisection, Newton-Raphson, regula falsi, secant methods, methods specific for polynomial roots finding	Exposition, debate, case studies, exercises	2, [2,3,4]
Interpolare numerica: Lagrange, forme Newton, spline, cele mai mici patrate Metode de optimizare numerica	Exposition, debate, case studies, exercises	2, [2, 3, 4]

Bibliography
1. G.H. Golub, C.F. Van Loan, Matrix Computations, JHU Press, 1996 2. Numerical analysis with algorithms and programming, Santanu Saha Ray, CRC Press, 2016, 3. R.L. Burden, J.D. Faires, Numerical Analysis– Brooks/Cole, Thomson Learning (10-th edition, 2015) 4. Nocedal, J. and Wright, S., Numerical Optimization, Springer Series in Operations Research and Financial Engineering (Springer New York, 2006)

8.2 Seminary / Laboratory	Teaching methods	Remarks (number oh hours, references)
Computing with errors, efficient algorithms for matrix multiplication	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,4]
LU Decompositions	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,2]
QR Decompositions	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,2,4]
Efficient storing methods for sparse matrices	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,2]
Iterative methods for solving linear systems of equations: Jacobi	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,2]
Iterative methods for solving linear systems of equations: Gauss-Seidel	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,2]
Iterative methods for solving linear systems of equations: methods for symmetric and positive definite matrices	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,2,3]
Nonlinear equations roots	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,2]
Approximating roots for polynomials	Descrierea problemei de rezolvat: algoritm si exemple	2, [1,2, 3]
Numerical Interpolation: Lagrange, Newton	Descrierea problemei de rezolvat: algoritm si exemple	2, [2, 3, 4]
Numerical Interpolation: spline functions, least squares solutions	Descrierea problemei de rezolvat: algoritm si exemple	2, [2, 3, 4]
Numerical Optimization	Descrierea problemei de rezolvat: algoritm si exemple	2, [2, 3, 4]

Bibliography
1. S. Salleh, A.Y. Zomaya, S.A. Bakar, Computing for Numerical Methods using Visual C++, Wiley-Interscience, 2008 2. D. E. Stewart, Numerical Analysis: A Graduate Course, Springer, 2022 3. Numerical Recipes, http://www.nr.com 4. R.L. Burden, J.D. Faires, Numerical Analysis– Brooks/Cole, Thomson Learning (10-th edition, 2015)

9. Coordination of the contents with the expectations of the community representatives, professional associations and relevant employers in the corresponding domain

The content of the course is designed to address the necessities of the employers from the IT industry.

10. Assessment and examination

10.1 Continuous assessment		Percentage (min. 30%)		100
Course	Assessment type			Written assessment
	Percentage			50
	Failure to pass the continuous assessment results in failure to pass the final assessment			Yes
	Assessment methods	Details	Percentage	with reexamination
		Test	100	Yes
Seminary / Laboratory	Assessment type			Oral assessment
	Percentage			50
	Failure to pass the continuous assessment results in failure to pass the final assessment			No
	Assessment methods	Details	Percentage	with reexamination
		Project	40	No
		Continuous oral assessment	60	No

10.2 Special notes (special situations is assessment)
The grade obtain at the course written test must be greater or equal to 3.

10.3 Minimum performance standard
Understanding the specific methods of Numerical Calculus. Knowing and being able to solve problems involving numerical methods. Has to have the skills of using Numerical Calculation libraries in solving real-life problems. The final score must be greater or equal to 45% of the maximal final score.

Date,

Course coordinator,
Assoc. Prof. PhD. ANDREEA-VALENTINA ARUSOAIE/ Assoc. Prof. PhD. ANCA IGNAT/ Assoc. Prof. PhD. ANCA IGNAT

Seminary coordinator,
Assoc. Prof. PhD. ANDREEA-VALENTINA ARUSOAIE/ Assoc. Prof. PhD. ANCA IGNAT/ Assoc. Prof. PhD. ANCA IGNAT/ Specialist PhD. SEBASTIAN-ADRIAN CIOBANU/ Specialist ROBERT-ADRIAN GAINA/ PhD Student RADU MIRON/ PhD Student IULIA-CATALINA PLESCA

Aproval date in the department,

Head of the departament,
Assoc. Prof. PhD. ANDREI ARUSOAIE